L-3 (IS)

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Def. Let a and b be integers with $a \neq 0$. a divides b, if there is an integer k such that $b = ak$. This is denoted by $a|b$. Means b is a multiple of a.

Ex $3|15$, $-15|60$, ~~7|18~~ $7 \nmid 18$

Proposition - Let a, b, c are integers

(i) For very $a \neq 0$, $a|0$ and $a|a$. Also $1|b$ for very b.

→ Since $0 = a \cdot 0$, $k = 0$ so $a|0$.

→ $a = a \cdot 1$ and $b = 1 \cdot b$

(ii) If $a|b$ and $b|c$ then $a|c$

→ $b = a \cdot k_1$ and $c = b \cdot k_2$ so $c = a \cdot (k_1 k_2)$

(iii) If $a|b$ and $a|c$ then $a|(sb + tc)$ for all integers s & t.

→ $b = ak_1$ and $c = a \cdot k_2$

So $sb + tc = sak_1 + tak_2 = a \cdot (sk_1 + tk_2)$

Prime Numbers -

A number $p > 1$ that is divisible only by 1 and itself is called a Prime number.

An integer ~~that is~~ $n > 1$ that is not Prime is called composite, it means n must be expressible as a product of integers.

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Greatest Common Divisor -

The greatest common divisor of a and b is the largest positive integer dividing both a and b and is denoted by $gcd(a,b)$.

Ex. $gcd(6,4) = 2$, $gcd(5,7) = 1$

a and b are relatively prime if $gcd(a,b) = 1$.

Method 1

$$576 = 2^6 \times 3^2, \quad 135 = 3^3 \times 5$$

$$\text{so } gcd(576,135) = 3^2$$

If a and b are large no then factoring of a and b is a time consuming process.

Method 2 (Using Euclideans algorithms)

gcd (482, 1180)

→ greater no → smaller no

$1180 = 2 \times 482 + 216$. Step-1

$482 = 2 \times 216 + 50$

$216 = 4 \times 50 + 16$

$50 = 3 \times 16 + 2$

$16 = 8 \times 2 + 0$ ← Last nonzero remainder.

The last nonzero remainder is the gcd.

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Euclidean algorithm

Let $a > b$

$$a = q_1 b + r_1$$

If $r_1 = 0$ then b divides a and gcd is b

If $r_1 \neq 0$ then continue

$$a = q_1 b + r_1$$

$$b = q_2 r_1 + r_2$$

$$r_1 = q_3 r_2 + r_3$$

$$r_2 = q_4 r_3 + r_4$$

$$\vdots$$

$$r_{k-2} = q_k \cdot r_{k-1} + r_k$$

$$r_{k-1} = q_{k+1} r_k + 0$$

So $\gcd(a, b) = r_k$.

It does not require factorization of numbers & it is fast.

Extended Euclidean algorithm

Given integers a and b, there are integers x and y such that

$$ax + by = \gcd(a, b)$$

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Find integers x and y such that

$\gcd(56, 72) = 56x + 72y$

$72 > 56$

$72 = 56 \times 1 + 16$

$56 = 16 \times 3 + 8$

$16 = 8 \times 2 + 0$

$\gcd(56, 72) = 8$

• $\gcd(a, b) = ax + by$

$\gcd(56, 72) = 8$

$8 = 56 - (16 \times 3)$

$= 56 - [(72 - 56 \times 1) \times 3]$

$= 56 - [72 \times 3 - 56 \times 3]$

$= 56 - 72 \times 3 + 56 \times 3$

$8 = 56 \times 4 - 72 \times 3$

$8 = ax + by$

So $x = 4, y = -3$

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Find integers x and y such that

$\gcd(272, 1479) = 272x + 1479y$

$1479 = 272 \times 5 + 119$

$272 = 119 \times 2 + 34$

$119 = 34 \times 3 + 17$

$34 = 17 \times 2 + 0$

$\gcd(272, 1479) = 17$

$\gcd(a, b) = ax + by$

$17 = 119 - 34 \times 3$

$= 119 - [(272 - 119 \times 2) \times 3]$

~~$= 119 - [\{272 - (1479 - 272 \times 5) \times 2\} \times 3]$~~

$= 119 - (272 \times 3 - 119 \times 6)$

$= 119 - 272 \times 3 + 119 \times 6$

$= 119 \times 7 - 272 \times 3$

$= (1479 - 272 \times 5) \times 7 - 272 \times 3$

$= 1479 \times 7 - 272 \times 35 - 272 \times 3$

$= 1479 \times 7 - 272 \times 38$

$= 1479 \times 7 + 272 \times (-38)$

~~So $x = 7, y = -38$~~ So $x = -38, y = 7$

Extended Euclidian Algorithm

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$r_0 = a$ $\quad$ $r_1 = b$

$s_0 = 1$ $\quad$ $s_1 = 0$

$t_0 = 0$ $\quad$ $t_1 = 1$

$\vdots$ $\quad$ $\vdots$

$r_{j+1} = r_{j-1} - q_j \cdot r_j$

$s_{j+1} = s_{j-1} - q_j \cdot s_j$

$t_{j+1} = t_{j-1} - q_j t_j$

computation stops when $r_{k+1} = 0$ and gives r_k as gcd of a and b.

Ex. Let $a = 240$, $b = 46$

$240 = 5 \times 46 + 10$

$46 = 4 \times 10 + 6$

$10 = 1 \times 6 + 4$

$6 = 1 \times 4 + 2$

$4 = 2 \times 2 + 0$

So $\gcd(a, b) = 2$

$\gcd(a, b) = r_k = a s_k + b t_k$

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i	q_{i-1}	r_i	s_i	t_i
0	-	240	1	0
1	-	46	0	1
2	$240 \div 46 = 5$	$240 - 5 \times 46 = 10$	$1 - 5 \times 0 = 1$	$0 - 5 \times 1 = -5$
3	$46 \div 10 = 4$	$46 - 4 \times 10 = 6$	$0 - 4 \times 1 = -4$	$1 - 4 \times -5 = 21$
4	$10 \div 6 = 1$	$10 - 1 \times 6 = 4$	$1 - 1 \times -4 = 5$	$-5 - 1 \times 21 = -26$
5	$6 \div 4 = 1$	$6 - 1 \times 4 = 2$	$-4 - 1 \times 5 = -9$	$21 - 1 \times -26 = 47$
6	$4 \div 2 = 2$	$4 - 2 \times 2 = 0$	$5 - 2 \times -9 = 23$	$-26 - 2 \times 47 = -120$

Hence

$$-9 \times 240 + 47 \times 46 = 2$$

$$ax + by = \gcd(a, b)$$

So inverse of 240 is -9.

inverse of 46 is 47.